

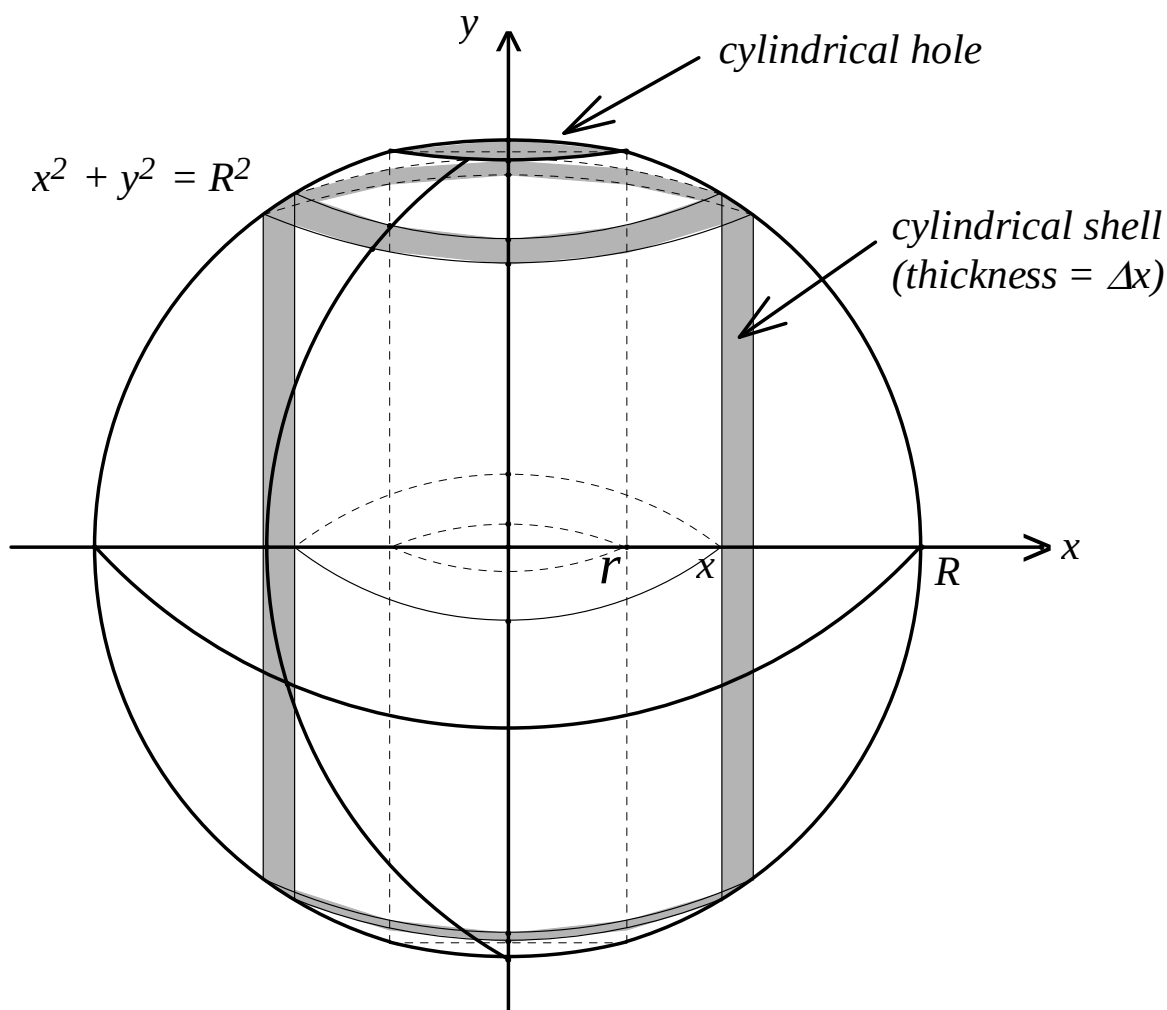
<u>QUESTION 1</u>	(15 Marks)	Marks
(a) (i)	Prove that the equation of the tangent to the rectangular hyperbola $xy = 16$ at the point $A\left(4p, \frac{4}{p}\right)$ is $x + p^2y = 8p$.	3
(ii)	If point B has co-ordinates $\left(4q, \frac{4}{q}\right)$, show that the equation of the chord AB is $x + pqy = 4(p + q)$.	2
(iii)	The tangent at A and the tangent at B intersect at T . Find the co-ordinates of T .	2
(iv)	If the chord AB passes through the point $R(8,8)$, show that the locus of T lies on the line $x + y = 4$.	2
(b)	A small object of 1 kg mass is fired vertically upwards from the ground with an initial speed of 50 ms^{-1} . At any instant the object is acted upon by gravity and a resistance of magnitude $\frac{1}{5}v$ where $v \text{ ms}^{-1}$ is the speed of the object at that instant. Taking the acceleration due to gravity as 10 ms^{-2} , prove that:	
(i)	the time for the object to reach its maximum height is $5\log_e 2$ seconds.	3
(ii)	the maximum height reached above the ground is $250(1 - \log_e 2)$ metres.	3

<u>QUESTION 2</u>	(START A NEW PAGE)	(16 Marks)	Marks
(a) (i)	A right-angled isosceles triangle has hypotenuse of length h units, show that its area equals $\frac{1}{4}h^2$ square units.		1
(ii)	A solid has for its base the area bounded by the parabola $y = 0.5x^2$ and the line $y = 3x$. Cross-sections of the solid are perpendicular to the base and parallel to the y -axis. Each cross-section is a right-angled isosceles triangle with its hypotenuse on the base. Find the volume of the solid.		3

QUESTION 2(b)(c) **(Continued on next page)**

QUESTION 2 (Continued)

- (b) A solid sphere of radius R has a cylindrical hole of radius r drilled right through it such that the axis of the hole passes through the centre of the sphere.



- (i) By taking cylindrical shells of width Δx and radius x , show that the volume of such a shell is approximately equal to $4\pi x\sqrt{R^2 - x^2} \Delta x$. 2
- (ii) Show that the volume of the remaining solid is $\frac{4}{3}\pi(R^2 - r^2)^{\frac{3}{2}}$. 3
- (c) A 1 kg mass moves in a straight line under the action of a constant driving force F Newtons. During this motion the mass encounters a resistive force of magnitude kv Newtons per unit mass where $v \text{ ms}^{-1}$ is its speed and k is a positive constant.
- (i) Explain why the acceleration, $\ddot{x} \text{ ms}^{-2}$, is given by $\ddot{x} = F - kv$ for some constant $k > 0$. 1
- (ii) Given that the mass is initially at the origin with a velocity of $u \text{ ms}^{-1}$, find a formula for its velocity, $v \text{ ms}^{-1}$, at time t seconds. 2
- (iii) If after T seconds the velocity of the mass is $2u \text{ ms}^{-1}$, show that $F = ku \left[\frac{2e^{kT} - 1}{e^{kT} - 1} \right]$. 2

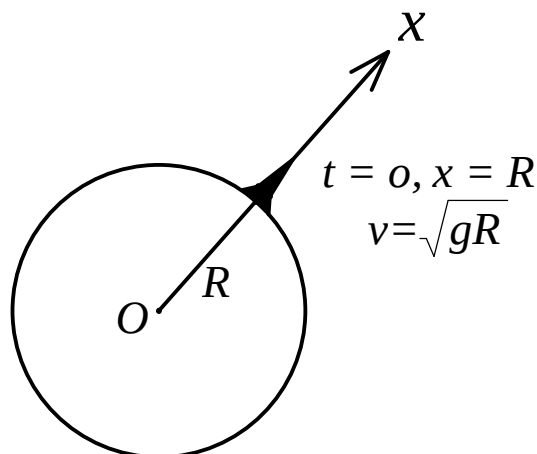
(iv) Prove that the distance, d metres, moved in this time is given by $d = \frac{FT - u}{k}$.

2

QUESTION 3**(START A NEW PAGE)****(16 Marks)****Marks**

- (a) (i) Prove that the equation the normal to the rectangular hyperbola $xy = c^2$ at the point $P\left(cp, \frac{c}{p}\right)$ is $p^2x - y = \frac{c}{p}(p^4 - 1)$. 1
 (You may assume that the equation of the tangent is $x + p^2y = 2cp$)
- (ii) The tangent at $P\left(cp, \frac{c}{p}\right)$ meets the x – axis at A and the y – axis at B . Find the coordinates of A and B and show that P is the midpoint of AB . 3
- (iii) The normal at $P\left(cp, \frac{c}{p}\right)$ meets the line $y = x$ at C and the line $y = -x$ at D . Find the coordinates of points C and D . 2
- (iv) What type of quadrilateral is $ACBD$ for all $p \neq 1$? (Give reasons). 2

- (b) A small rocket is projected vertically upwards from the Earth's surface with an initial speed of $\sqrt{gR}$ ms⁻¹. You may assume that the acceleration of an object at a distance x metres from the centre of the Earth is of magnitude $\frac{gR^2}{x^2}$, and directed towards the Earth's Centre, where R metres is the radius of the Earth.



- (i) Neglecting air resistance, show that the speed v ms⁻¹ of the rocket at distance x metres from the Earth's centre is given by $v = \sqrt{gR} \sqrt{\frac{2R - x}{x}}$. 3
- (ii) Find the maximum height that the rocket will reach above the Earth's surface. 1
- (iii) Find the time required to reach a height of R metres above the Earth's surface. 4

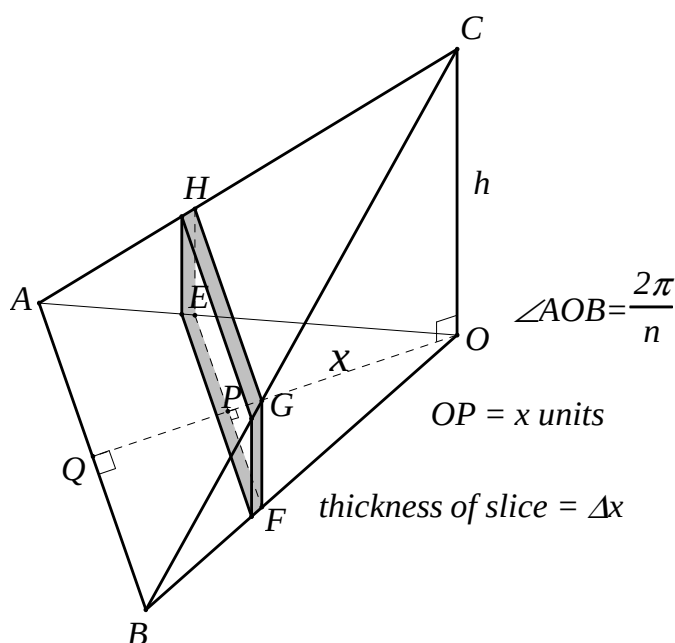
QUESTION 4**(START A NEW PAGE)****(15 Marks)****Marks**

- (a) A plane of mass M kg lands with velocity $u \text{ ms}^{-1}$ on a level airstrip. Upon landing it experiences a resistive force of magnitude αv^2 Newtons due to air resistance and braking, and a constant resistive force of magnitude β Newtons due to the friction between the tyres and the ground where α and β are constants.

(i) Show that the distance required to bring the plane to rest is $\frac{M}{2\alpha} \ln \left(1 + \frac{\alpha}{\beta} u^2 \right)$ metres. 3

(ii) Show that the time required for the plane to come to rest is $\frac{M}{\sqrt{\alpha\beta}} \tan^{-1} \left(u \sqrt{\frac{\alpha}{\beta}} \right)$ seconds. 3

- (b) OAB is an isosceles triangle with $OA = OB = r$ and $\angle AOB = \frac{2\pi}{n}$. $OABC$ is a triangular pyramid with $OC = h$ and OC is perpendicular to the plane AOB .



Consider a slice with cross-section $EFGH$ perpendicular to the plane AOB with $EF \parallel AB$, thickness Δx and at a perpendicular distance x units along OQ from the point O .

(i) Show that the volume of the slice is given by $\left(2h \tan \frac{\pi}{n} \right) \left(x - \frac{x^2}{r \cos \frac{\pi}{n}} \right) \Delta x$ 3

(ii) Hence show that the volume of the pyramid is $\frac{hr^2}{6} \sin \frac{2\pi}{n}$. 4

- (iii) Suppose that n identical pyramids $OABC$ are arranged about a common vertical axis OC to form a solid. Find the limit value of the volume as n becomes very large. 2

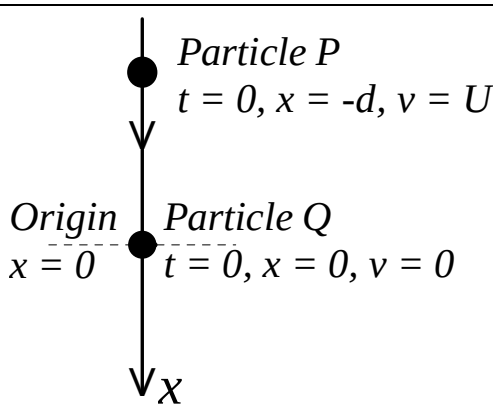
(You may assume that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$)



THIS IS THE END OF THE EXAMINATION



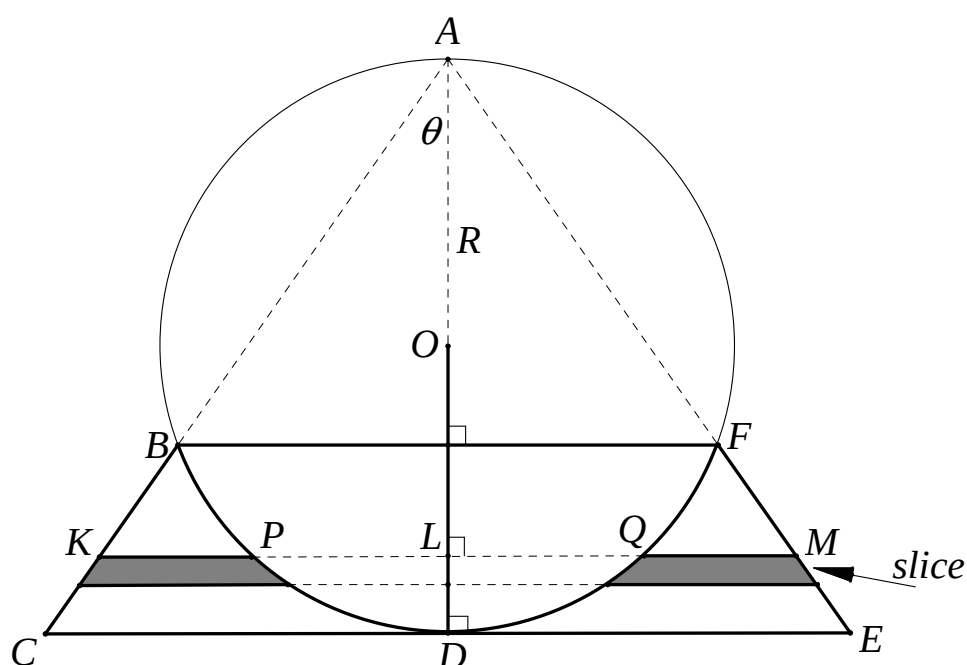
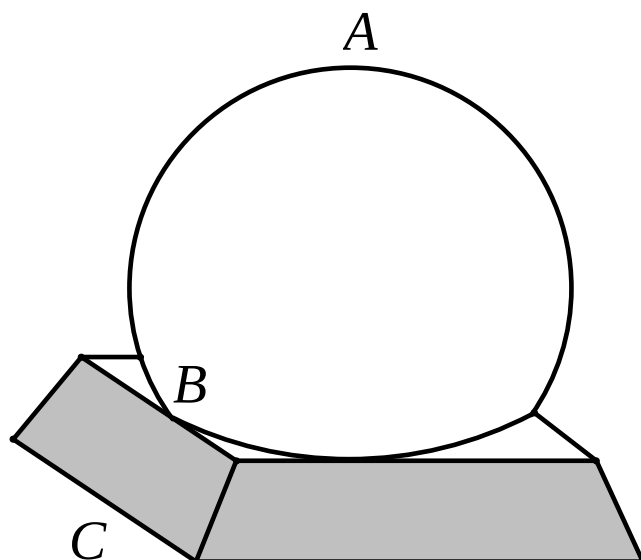
ADDITIONAL QUESTIONS

A particle P is thrown vertically downwards in a medium where the resistive force is proportional to the speed $v \text{ ms}^{-1}$ of the particle. The initial speed of particle P is $U \text{ ms}^{-1}$ and the particle is thrown from a point d metres above the origin and the acceleration due to gravity is $g \text{ ms}^{-2}$.			Marks
(i)	Explain why the acceleration, $\ddot{x} \text{ ms}^{-2}$, is given by $\ddot{x} = g - kv$ for some constant $k > 0$.		1
(ii)	Show that $v = \frac{g}{k} - \left(\frac{g - kU}{k} \right) e^{-kt}$.		2
(iii)	Show that $x = \frac{gt - kd}{k} - \left(\frac{g - kU}{k^2} \right) (e^{-kt} - 1)$.		2
(iv)	A second identical particle Q is dropped from the origin at the same instant that P is thrown down. Using the above results to write down similar expressions for the velocity and displacement of particle Q .		2
(v)	Find when particles P and Q collide and the speed with which they collide.		

			Marks
(i)	By considering areas or using integration techniques show that $\int_0^{\frac{a}{\sqrt{2}}} \sqrt{a^2 - x^2} \, dx = \frac{a^2(\pi + 2)}{8}.$		2
(ii)	The area of the minor segment bounded by the chord $x = \frac{a}{\sqrt{2}}$ and the circle $x^2 + y^2 = a^2$ is rotated one revolution about the chord. By considering circular cross-sections perpendicular to the chord, find the volume of the solid formed.		4

Mr Dud's crystal ball rests on a solid stand which is in the shape of a square based frustrum as shown.

Marks

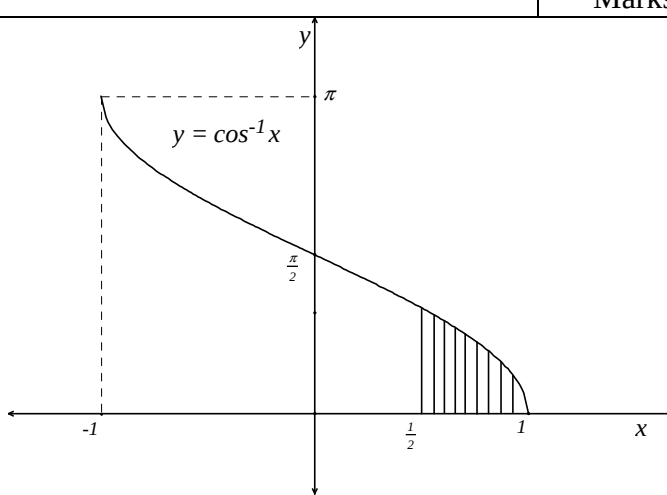


cross-section

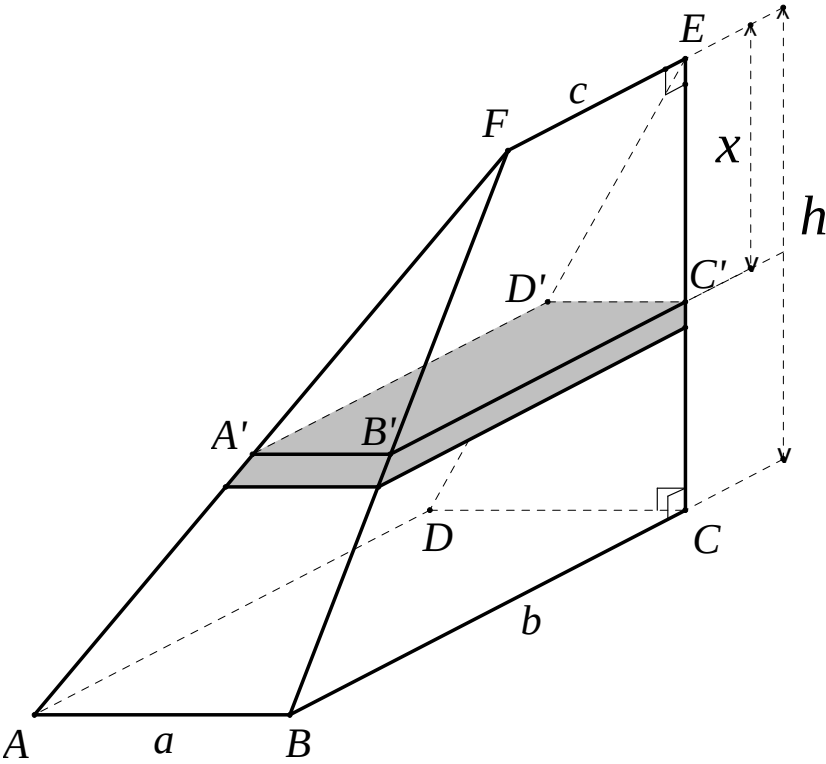
The stand is constructed so that the crystal ball of radius R fits snugly inside and just touches the centre of the square base. The side BC of the base slopes so that if extended it would pass through the top-most point of the ball at A and makes an angle θ with the vertical AD . Letting the distance OL be x units.

(i)	Explain why $LQ = \sqrt{R^2 - x^2}$ and $LM = (R + x)\tan \theta$	2
(ii)	Consider a slice KLM of thickness Δx as shown perpendicular to AD . Show that it has a volume ΔV units ³ given by $\Delta V \approx \left[4 \tan^2 \theta (R + x)^2 - \pi (R^2 - x^2) \right] \Delta x$	2
(iii)	Find the volume of such a solid when the angle $\theta = \frac{\pi}{6}$.	3

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		Marks
The region bounded by the curve $y = \cos^{-1} x$, $x = \frac{1}{2}$ and the x – axis is rotated one revolution about the y – axis to generate a solid.		
(i)	Show that an annular cross-section of the solid, parallel to the x – axis with height Δy has its volume given by $\Delta V \approx \pi \left(\cos^2 y - \frac{1}{4} \right) \Delta y$.	2
(ii)	Hence find the volume of the solid.	3

Consider solid $ABCDEF$ whose height $EC = h$ and whose base is a rectangle $ABCD$ with $AB = a$, $BC = b$ and top edge $EF = c$.	Marks
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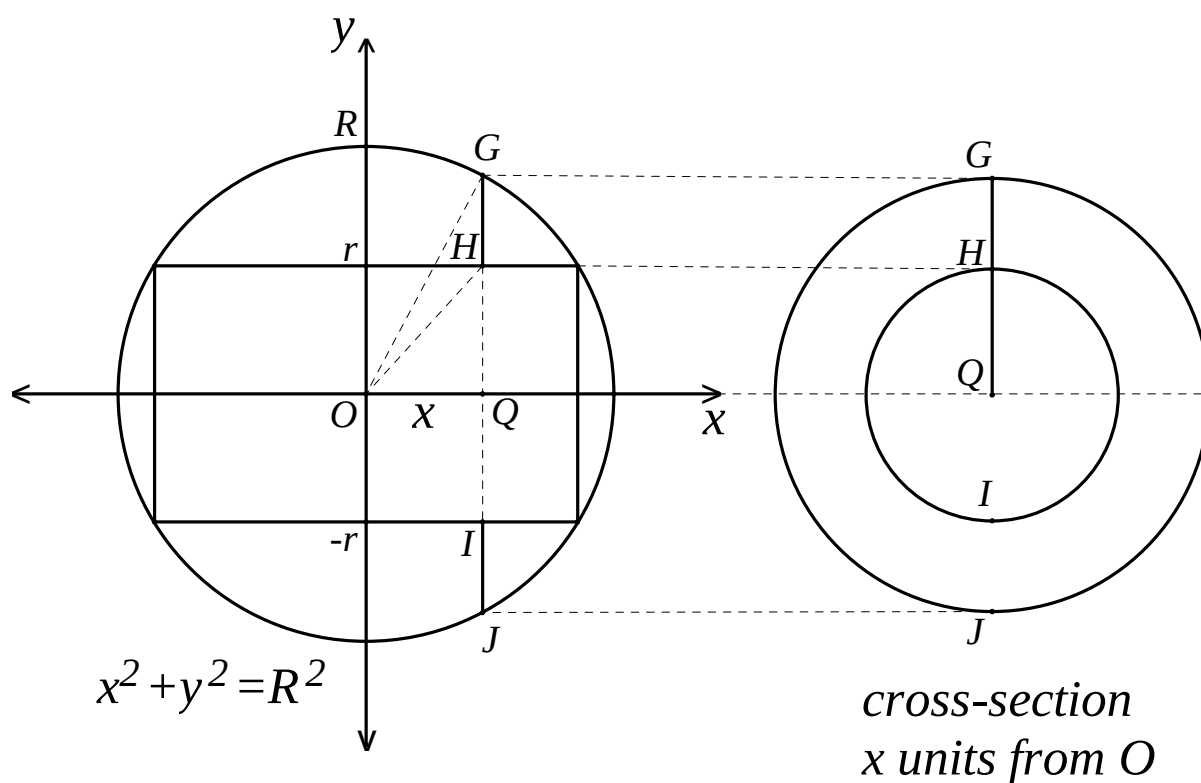
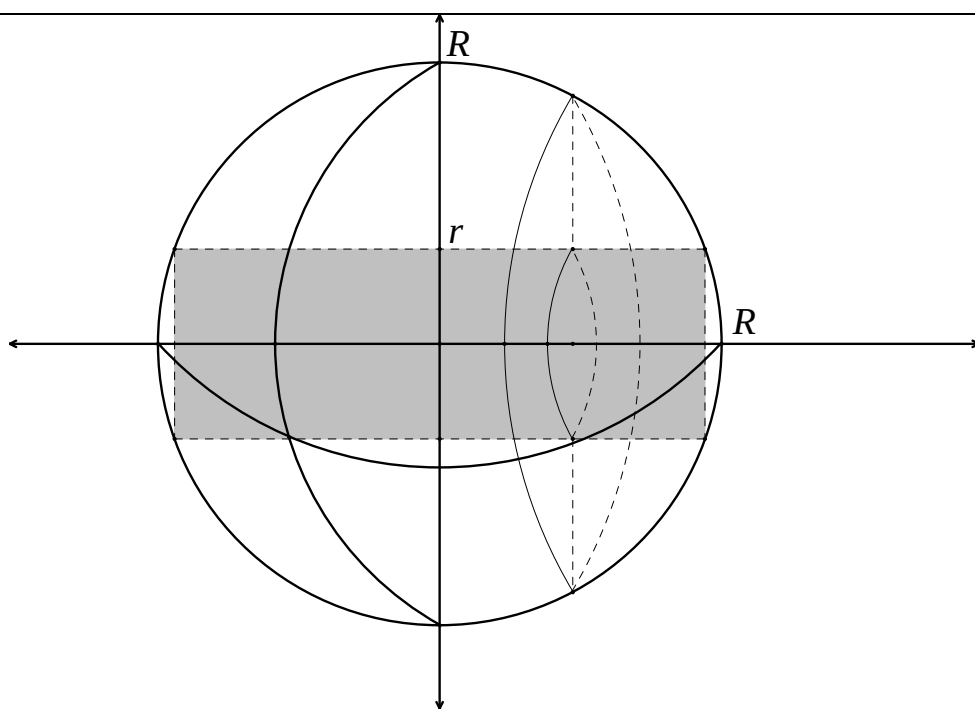


Consider a rectangular slice $A'B'C'D'$ (parallel to the base $ABCD$) x units from the top edge and with thickness Δx units.	
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(i)	Show that the volume of the slice is given by $\Delta V = \frac{ax}{b} \left(c + \frac{b-c}{b} x \right) \Delta x$.	
(ii)	Hence prove that the volume of the solid is $\frac{1}{6} ha(2b + c)$.	

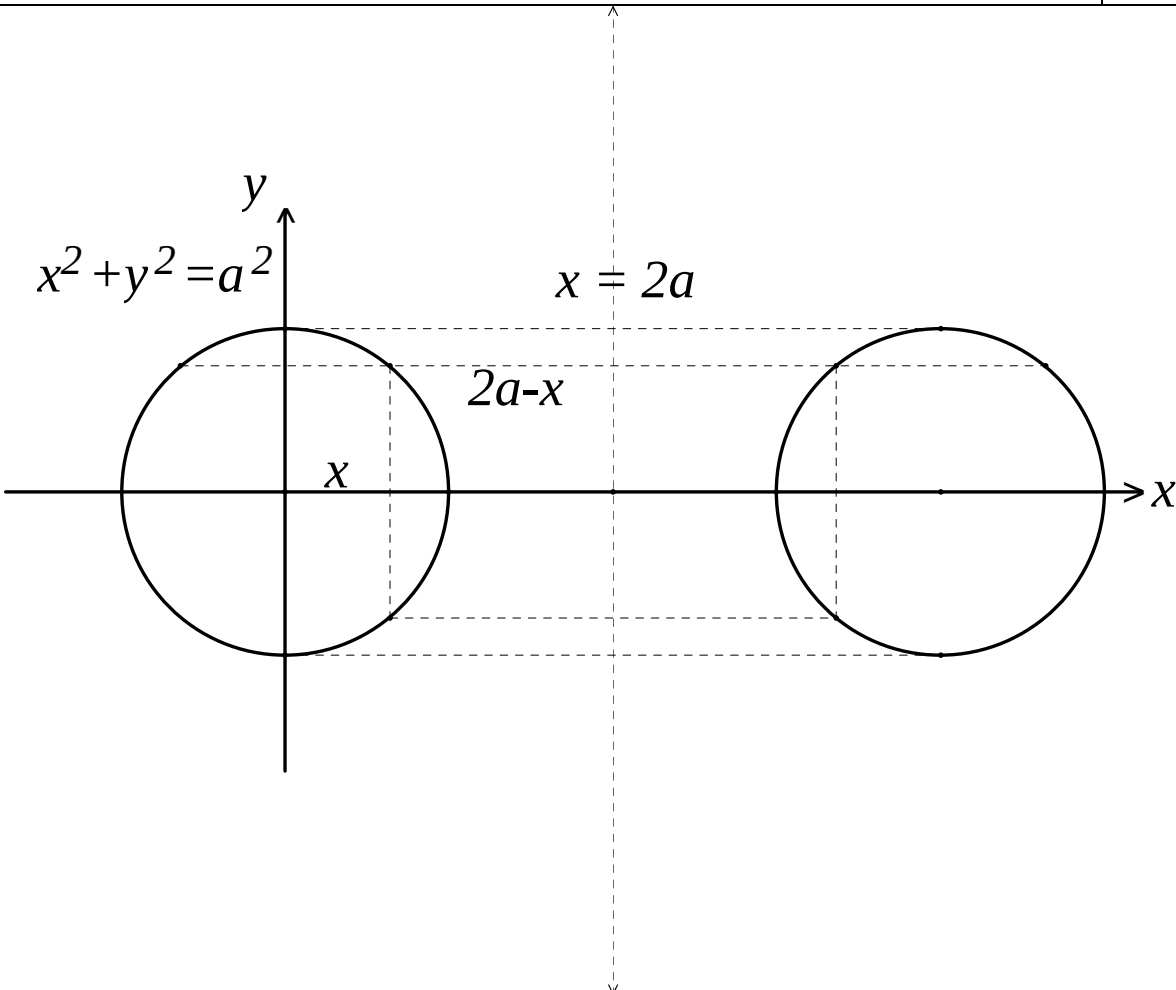
A solid sphere of radius R has a cylindrical hole of radius r drilled right through it such that the axis of the hole passes through the centre of the sphere.

Marks



(i) By considering annular cross-sections perpendicular to the x -axis and distance x units from the origin, show that the area of the annular cross-section is $\pi(R^2 - r^2 - x^2)$.

(ii) Show that the volume of the remaining solid is $\frac{4}{3}\pi(R^2 - r^2)^{\frac{3}{2}}$.

The circle $x^2 + y^2 = a^2$ is rotated one revolution about the line $x = 2a$ to form a torus.		Marks
		
(i)	By taking cylindrical shells of width Δx and axis $x = 2a$, show that the volume of such a shell is approximately equal to $4\pi(2a - x)\sqrt{a^2 - x^2} \Delta x$.	
(ii)	Hence show that the volume of the torus is $4\pi^2 a^3$ units ³ .	

The normal at a variable point $P\left(2p, \frac{2}{p}\right)$ on the hyperbola $xy = 4$ meets the x -axis at Q.		Marks
(i)	Prove that the normal at P is $p^2x - y = \frac{2}{p}(p^4 - 1)$.	2
(ii)	Find the co-ordinates of P.	1
(iii)	Find the co-ordinates of M, the midpoint of PQ.	1
(iv)	Hence show that the locus of M lies on the curve $y^4 + xy = 2$.	2

TERM 2 2010	EXT 2	MATHEMATICS: Question ...	(1)
Suggested Solutions		Marks	Marker's Comments
a) i) $y = \frac{16}{x} \quad (x \neq 0)$ $\therefore \frac{dy}{dx} = -\frac{16}{x^2}$ $= -\frac{16}{16p^2} = -\frac{1}{p^2} \text{ at } A$ $\therefore \text{Tangent at } A \text{ is } y - \frac{4}{p} = -\frac{1}{p^2}(x - 4p)$ $p^2y - 4p = -x + 4p$ $x + p^2y = 8p$		1 1 1	
ii) Two point form: $\frac{y - \frac{4}{p}}{\frac{4}{q} - \frac{4}{p}} = \frac{x - 4p}{4q - 4p}$ $\frac{pqy - 4q}{4p - 4q} = \frac{x - 4p}{4q - 4p}$ $pqy - 4q = -(x - 4p) \quad (p \neq q)$ $x + pqy = 4(p + q)$		2	
iii) Tangent at B is $x + q^2y = 8q$ ① Solve at T $x + p^2y = 8p$ ② ① - ② $(q^2 - p^2)y = 8(q - p)$ $(q + p)y = 8 \quad (p \neq q)$ $y = \frac{8}{p + q}$ Substitute into ① $x = 8q - \frac{8q^2}{p + q}$ $= \frac{8pq + 8q^2 - 8q^2}{p + q}$ $= \frac{8pq}{p + q}$ $\therefore T \text{ is } \left(\frac{8pq}{p + q}, \frac{8}{p + q} \right)$		1 1	$\frac{1}{2}$ off if $p \neq q$ not mentioned at all. $\frac{1}{2}$ off if T not gathered together at the end.

Suggested Solutions

Marks

Marker's Comments

iv) The chord in (ii) goes through (8,8)

$$\therefore 8 + 8pq = 4(p+q)$$

$$\underline{2(1+pq) = (p+q)}$$

Substitute T into $x+y=4$

$$\text{LHS } x+y = \frac{8pq}{p+q} + \frac{8}{p+q}$$

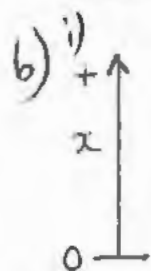
$$= \frac{8(pq+1)}{p+q}$$

$$= \frac{8(p+q)/2}{p+q}$$

$$= 4 \quad (p \neq -q)$$

$$= \text{RHS}$$

$\therefore T$ lies on the line $x+y=4$



FORCES

ACCNS.



$$m\ddot{x} = -mg - \frac{v}{5} \quad (\text{Newton's 2nd Law})$$

$$\text{But } m=1, g=10$$

$$\therefore \ddot{x} = \frac{dv}{dt} = -10 - \frac{v}{5}$$

$$\frac{dv}{dt} = \frac{50+v}{5}$$

$$\therefore \int_{50}^0 \frac{dv}{50+v} = \int_0^T \frac{-dt}{5} \quad (\text{Where } T \text{ is the required time})$$

$$\therefore [\ln(50+v)]_{50}^0 = \left[-\frac{t}{5}\right]_0^T$$

Diagrams were poorly done. For a given result, something was expected.

MATHEMATICS Extension 2: Question

(3)

Suggested Solutions

Marks

Marker's Comments

$$\ln 50 - \ln 100 = -\frac{T}{5}$$

$$\ln \frac{50}{100} = -\frac{T}{5}$$

$$\ln \frac{100}{50} = \frac{T}{5}$$

$$\therefore T = 5 \ln 2$$

1

Time to zero velocity is $5 \ln 2$ secs

ii) $v \frac{dv}{dx} = -10 - \frac{v}{5}$ (different from Q i)

1

$$\int_{50}^0 \frac{v dv}{50+v} = - \int_0^H \frac{dx}{5} \quad \text{where } H \text{ is the maximum height}$$

$$\int_{50}^0 \frac{(50+v) - 50}{(50+v)} dv = - \left[\frac{x}{5} \right]_0^H$$

1

$$\left[v - 50 \ln(50+v) \right]_{50}^0 = -\frac{H}{5}$$

$$-50 \ln 50 - 50 + 50 \ln 100 = -\frac{H}{5}$$

$$50 \ln \left(\frac{100}{50} \right) - 50 = -\frac{H}{5}$$

$$\therefore H = 5(50 - 50 \ln 2)$$

$$= 250(1 - \ln 2)$$

1

Particle reaches maximum height of

$250(1 - \ln 2)$ metres

MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

Marker's Comments

2e(1)

$$\begin{array}{c} F \rightarrow \\ \leftarrow mkv \\ \hline \rightarrow +k \end{array}$$

$$2e(1) \quad \frac{d}{dt} = F - mkv$$

$$mk = F - mkv$$

$m=1$

$$v'' = F - kv$$

(ii)

$$\frac{dv}{dt} = F - kv$$

$$\int_0^t dt = \int_u^v \frac{dv}{F - kv}$$

$$t = \frac{1}{k} \ln \left(\frac{F - km}{F - kv} \right)$$

$$v = \frac{1}{k} \left(F - (F - km)e^{-kt} \right)$$

(iii)

$$T = \frac{1}{k} \ln \left(\frac{F - km}{F - 2km} \right)$$

$$kT = \ln \left(\frac{F - km}{F - 2km} \right)$$

$$F - km = F e^{kT} - 2km e^{kT}$$

$$F = km \left[\frac{2e^{kT} - 1}{e^{kT} - 1} \right]$$

(iv)

$$v \frac{dv}{du} = F - kv$$

$$\int_0^u du = \int_u^{2u} \frac{v dv}{F - kv}$$

$$u = -\frac{1}{k} \left[u + \frac{F}{k} \ln \left(\frac{F - 2ku}{F - ku} \right) \right]$$

Substitute $v=2u$ into (ii) $T = \frac{1}{k} \ln \left(\frac{F - ku}{F - 2ku} \right)$

$$\therefore u = -\frac{1}{k} \left[u - F \cdot T \right]$$

$$\text{distance } d = \frac{FT - u}{k}$$

1

Conclusion e^{-kt}
 $kv \geq mkv$

1 1/2

1/2

difficulty students
writing u & v

need
qualified
 $F > kv$

2

1/2 off each
algebra error

1

1/2

1/2

T requires
justification

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

(a) (i) Equation of tangent $x + p^2 y = 2cp$

$$m_T = -\frac{1}{p^2}$$

$$m_N = p^2$$

Equation of normal:
 $y - y_1 = m(x - x_1)$

at $P(cp, \frac{c}{p})$

$$y - \frac{c}{p} = p^2(x - cp)$$

$$p^2 x - y = cp^3 - \frac{c}{p}$$

$$\therefore p^2 x - y = \frac{c}{p}(p^4 - 1)$$

Alternatively

Equation of perpendicular line to tangent is

$$p^2 x - y = k$$

Sub $P(cp, \frac{c}{p})$

$$p^2(cp) - \frac{c}{p} = k$$

$$\therefore p^2 x - y = cp^3 - \frac{c}{p}$$

$$p^2 x - y = \frac{c}{p}(p^4 - 1)$$

(ii) at A $y = 0$ tangent $x + p^2 y = 2cp$

$$x = 2cp \quad A(2cp, 0)$$

at B $x = 0$ $p^2 y = 2cp$

$$y = \frac{2c}{p} \quad B(0, \frac{2c}{p})$$

①

③

① A

① B

some working must be shown

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

midpoint of AB

$$x_c = \frac{2cp + 0}{2} = cp$$

$$y = \frac{0 + \frac{2c}{p}}{2} = \frac{c}{p}$$

$\therefore P$ is midpoint of AB.

(ii) Normal $p^2x - y = \frac{c}{p}(p^4 - 1)$

at C $y = x$

$$p^2x - x = \frac{c}{p}(p^4 - 1)$$

$$x(p^2 - 1) = \frac{c}{p}(p^2 - 1)(p^2 + 1)$$

$$x = \frac{c}{p}(p^2 + 1) \quad p^2 - 1 \neq 0$$

$$\therefore y = \frac{c}{p}(p^2 + 1)$$

$$C \left[\frac{c}{p}(p^2 + 1), \frac{c}{p}(p^2 + 1) \right]$$

at D $y = -x$

Normal $p^2x - y = \frac{c}{p}(p^4 - 1)$

$$p^2x + x = \frac{c}{p}(p^4 - 1)$$

$$x(p^2 + 1) = \frac{c}{p}(p^2 - 1)(p^2 + 1)$$

$$x = \frac{c}{p}(p^2 - 1)$$

$$y = -\frac{c}{p}(p^2 - 1) = \frac{c}{p}(1 - p^2)$$

$$D \left[\frac{c}{p}(p^2 - 1), \frac{c}{p}(1 - p^2) \right]$$

① Proof

must show working to calculate midpoint AB coordinates

②

② for x

② for y

no marks lost if $p^2 - 1 \neq 0$ omitted

② for x

② for y

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

(iv)

Find midpoint of CD

$$x = \left[\frac{c}{p}(p^2+1) + \frac{c}{p}(p^2-1) \right] \div 2$$

$$= \frac{c}{p} \left[\frac{2p^2}{2} \right] = cp$$

$$y = \left[\frac{c}{p}(p^2+1) + \frac{c}{p}(1-p^2) \right] \div 2$$

$$= \frac{c}{p} \left[\frac{2}{2} \right] = \frac{c}{p}$$

 $\therefore P$ is midpoint of CD. $\therefore$ Diagonals bisect each other
ACBD is a parallelogram.AS AB is on the tangent
CD is on the normal. $\therefore$ AB is perpendicular to CD.ACBD is a rhombus
(diagonals bisect at right angles)

TEST angle between sides BC and AC

$$m_{BC} = \frac{\frac{c}{p}(p^2+1) - 2\frac{c}{p}}{\frac{c}{p}(p^2+1) - 0}$$

$$= \frac{\frac{c}{p}(p^2+1-2)}{\frac{c}{p}(p^2+1)} = \frac{p^2-1}{p^2+1}$$

$$m_{AC} = \frac{\frac{c}{p}(p^2+1)}{\frac{c}{p}(p^2+1) - 2cp}$$

$$= \frac{\frac{c}{p}(p^2+1)}{\frac{c}{p}(p^2+1-2p^2)} = \frac{p^2+1}{1-p^2}$$

 $\therefore m_{BC} \times m_{AC} = -1 \therefore BC$ is perpendicular to AC. $\therefore$ ACBD is a square. Rhombus with one right angle.

If correct reasoning given

⑤ kite

② parallelogram

① rhombus

① rectangle

② square

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

$$3(b) i \quad \frac{d(\frac{1}{2}v^2)}{dx} = -\frac{gR^2}{x^2}$$

$$\int_{\sqrt{gR}}^v d(\frac{1}{2}v^2) = -gR^2 \int_R^x \frac{1}{x^2} dx$$

① integral

$$\left[\frac{1}{2}v^2 \right]_{\sqrt{gR}}^v = -gR^2 \left[-\frac{1}{x} \right]_R^x$$

② integrating

$$\frac{1}{2}v^2 - \frac{1}{2}gR = gR^2 \left[\frac{1}{x} - \frac{1}{R} \right]$$

③ sub limits (or finding c)

$$v^2 = gR^2 \left[\frac{2(R-x)}{Rx} \right] + gR$$

$$v^2 = \frac{gR^2 [2R - 2x + x]}{Rx}$$

$$= gR^2 \left[\frac{2R - x}{Rx} \right]$$

④ simplifying

$$v = \sqrt{gR \left(\frac{2R-x}{x} \right)} \quad \text{as } v > 0 \text{ for speed.}$$

⑤ $v > 0$

(ii) maximum height occurs when $v = 0$

$$0 = \sqrt{gR \left(\frac{2R-x}{x} \right)}$$

$$\therefore x = 2R$$

⑥ $x = 2R$

max height above Earth

$$= 2R - R$$

$$= R \text{ metres}$$

⑦ $H = R$

3

MATHEMATICS Extension 2: Question...

Suggested Solutions

Marks

Marker's Comments

$$3(b)(iii) v = \sqrt{gR} \sqrt{2R-x}$$

$$\frac{dx}{dt} = \sqrt{gR} \sqrt{2R-x}$$

$$\int_R^{2R} \frac{dx}{\sqrt{2R-x}} = \sqrt{gR} \int_0^T dt$$

(4)

① integral

Method ①

$$\int_R^{2R} \frac{x}{\sqrt{2Rx-x^2}} dx = \sqrt{gR} T$$

$$\therefore \sqrt{gR} T = \frac{1}{2} \int_R^{2R} \left(\frac{2x-2R}{\sqrt{2Rx-x^2}} + \frac{2R}{\sqrt{2Rx-x^2}} \right) dx$$

$$= \left[-\frac{1}{2} \times 2 \sqrt{2Rx-x^2} \right]_R^{2R} + R \sin^{-1} \left[\frac{x-R}{R} \right]_R^{2R}$$

① separation of fraction

① integration

$$= [0 + R \sin^{-1}(1)] - [-\sqrt{R^2} - 0]$$

$$= R \left(\frac{\pi}{2} + 1 \right)$$

① answer

$$\therefore \text{Time} = \frac{\sqrt{R}}{\sqrt{g}} \left[\frac{\pi}{2} + 1 \right]$$

MATHEMATICS Extension 2: Question 3

Suggested Solutions

Marks

Marker's Comments

3(b)(iii) Alternative method (2)

$$\sqrt{gR} \int_0^T dt = \int_R^{2R} \frac{\sqrt{x}}{\sqrt{2R-x}} dx$$

$$\text{Let } x = 2R \sin^2 \theta$$

$$x = 2R \quad \theta = \frac{\pi}{2}$$

$$x = R \quad \theta = \frac{\pi}{4}$$

$$\frac{dx}{d\theta} = 4R \sin \theta \cos \theta$$

$$\sqrt{gR} T = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sqrt{2R \sin^2 \theta} \times 4R \sin \theta \cos \theta d\theta}{\sqrt{2R - 2R \sin^2 \theta}}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sin^2 \theta \times 4R \sin \theta \cos \theta d\theta}{\cos^2 \theta}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4R \tan \theta \sin \theta \cos \theta d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4R \sin^2 \theta d\theta$$

$$= 4R \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= 2R \left[\theta - \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= 2R \left\{ \left[\frac{\pi}{2} - 0 \right] - \left[\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right] \right\}$$

$$= R \left[\frac{\pi}{2} + 1 \right]$$

$$\text{Time} = \frac{1}{\sqrt{g}} \left[\frac{\pi}{2} + 1 \right]$$

(4)

① integral

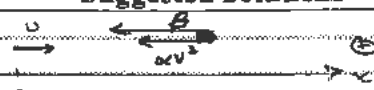
① correct substitution and limits

① integration

① answer.

M. EXT 2 TERM 2 ASSESSMENT, 2010
MARKING SCHEME

MATHEMATICS Extension 2: Question... **4**

Q4	Suggested Solutions	Marks	Marker's Comments
(a) (i)	 Let $t=0$ on landing, $x=0$, $v=u$ Let $t=T$, $x=D$ when $v=0$ FORCE EQU: $M\ddot{x} = -\beta - \alpha v^2$ (Newton's II) ✓ $\ddot{x} = -\frac{1}{M}(\beta + \alpha v^2)$ $\frac{d}{dx} \frac{1}{2} v^2 = v \frac{dv}{dx} = -\frac{1}{M}(\beta + \alpha v^2)$ $\therefore \frac{v dv}{\beta + \alpha v^2} = -\frac{1}{M} dx$ so $\int_u^0 \frac{v dv}{\beta + \alpha v^2} = -\frac{1}{M} \int_0^D dx$ i.e. $\frac{1}{2\alpha} [\ln(\beta + \alpha v^2)]_u^0 = -\frac{1}{M} [x]_0^D = -\frac{D}{M}$ ✓ $\therefore -\frac{D}{M} = \frac{1}{2\alpha} [\ln \beta - \ln(\beta + \alpha u^2)]$ $\therefore D = \frac{M}{2\alpha} \left[\ln \left(\frac{\beta + \alpha u^2}{\beta} \right) \right]$ ✓ Distance travelled = $\frac{M}{2\alpha} \ln \left(1 + \frac{\alpha u^2}{\beta} \right)$ m. qed.	1	$\frac{1}{2\alpha} \ln(\beta + \alpha v^2) = -\frac{x}{M} + C$ $v=0 \Rightarrow v=u$ $C = \frac{1}{2\alpha} \ln(\beta + \alpha u^2)$ etc.
(ii)	$\ddot{x} = \frac{dv}{dt} = -\frac{1}{M}(\beta + \alpha v^2)$ $\frac{dv}{\beta + \alpha v^2} = -\frac{1}{M} dt$ so $\int_u^0 \frac{dv}{\beta + \alpha v^2} = -\frac{1}{M} \int_0^T dt$ ✓ $\frac{1}{\alpha} \int_u^0 \frac{dv}{\frac{\beta}{\alpha} + v^2} = -\frac{1}{M} [t]_0^T = -\frac{T}{M}$ $\frac{1}{\alpha} \cdot \frac{1}{\sqrt{\frac{\beta}{\alpha}}} \left[\tan^{-1} \frac{v}{\sqrt{\frac{\beta}{\alpha}}} \right]_u^0 = -\frac{T}{M}$ i.e. $\frac{1}{\sqrt{\alpha\beta}} \left[\tan^{-1} \left(v \sqrt{\frac{\alpha}{\beta}} \right) \right]_u^0 = -\frac{T}{M}$ so $-\frac{T}{M} = \frac{1}{\sqrt{\alpha\beta}} \left[\tan^{-1}(0) - \tan^{-1} \left(u \sqrt{\frac{\alpha}{\beta}} \right) \right]$ ✓ $\therefore$ time taken = $\frac{M}{\sqrt{\alpha\beta}} \tan^{-1} \left(u \sqrt{\frac{\alpha}{\beta}} \right)$ seconds. qed.	1	$\frac{1}{\alpha} \cdot \frac{1}{\sqrt{\frac{\beta}{\alpha}}} \tan^{-1} \frac{v}{\sqrt{\frac{\beta}{\alpha}}} = -\frac{t}{M} + C$ $t=0 \Rightarrow v=u$ $\therefore C = \frac{1}{\sqrt{\alpha\beta}} \tan^{-1} u \sqrt{\frac{\alpha}{\beta}}$

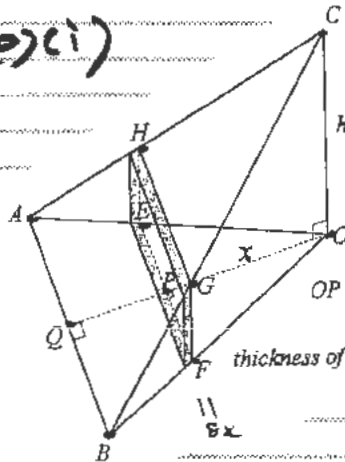
MATHEMATICS Extension 2: Question 4

Suggested Solutions

Marks

Marker's Comments

(b)(i)



$$OA = OB = r$$

TOP VIEW

Isos ΔAOB



$$\tan \frac{\pi}{n} = \frac{PF}{x}$$

$$PF = x \tan \frac{\pi}{n}$$

$$\therefore EF = 2x \tan \frac{\pi}{n}$$

$$\cos \frac{\pi}{n} = \frac{OQ}{r}; \sin \frac{\pi}{n} = \frac{QB}{r}$$

$$\therefore OQ = r \cos \frac{\pi}{n}; QB = r \sin \frac{\pi}{n}$$

$$AB = 2r \sin \frac{\pi}{n}$$

$\Delta POF \parallel \Delta QOB$ (equiangular)

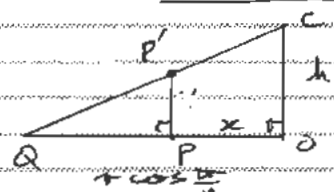
$\therefore \frac{PF}{QB} = \frac{OP}{OQ}$ (corresp. sides in similar triangles are in same ratio)

$$\frac{PF}{r \sin \frac{\pi}{n}} = \frac{x}{r \cos \frac{\pi}{n}}$$

$$PF = x \tan \frac{\pi}{n}$$

$$\therefore EF = 2x \tan \frac{\pi}{n}$$

SIDE VIEW



$\Delta QPP' \parallel \Delta QOC$ (equiangular)

$\therefore \frac{PP'}{h} = \frac{QP}{QO}$ (corresp. sides in similar triangles are in same ratio)

$$= \frac{QO - x}{QO} = 1 - \frac{x}{QO}$$

$$\therefore PP' = h \left(1 - \frac{x}{r \cos \frac{\pi}{n}} \right)$$

$$\therefore \delta V = EF \times PP' \times \delta x$$

$$\text{i.e. } \delta V = 2x \tan \frac{\pi}{n} \times h \left(1 - \frac{x}{r \cos \frac{\pi}{n}} \right) \delta x$$

$$\delta V = 2h \tan \frac{\pi}{n} \left(x - \frac{x^2}{r \cos \frac{\pi}{n}} \right) \delta x \quad \text{q.e.d.}$$

(ii)

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{r \cos \frac{\pi}{n}} 2h \tan \frac{\pi}{n} \left(x - \frac{x^2}{r \cos \frac{\pi}{n}} \right) \delta x$$

$$= 2h \tan \frac{\pi}{n} \int_0^{r \cos \frac{\pi}{n}} \left(x - \frac{x^2}{r \cos \frac{\pi}{n}} \right) dx$$

$$= 2h \tan \frac{\pi}{n} \left[\frac{1}{2} x^2 - \frac{x^3}{3r \cos \frac{\pi}{n}} \right]_0^{r \cos \frac{\pi}{n}}$$

$$= 2h \tan \frac{\pi}{n} \left[\left(\frac{r^2 \cos^2 \frac{\pi}{n}}{2} - \frac{r^3 \cos^3 \frac{\pi}{n}}{3r \cos \frac{\pi}{n}} \right) - 0 \right]$$

$$= 2h \tan \frac{\pi}{n} \times \left[\frac{r^2 \cos^2 \frac{\pi}{n}}{2} - \frac{r^2 \cos^2 \frac{\pi}{n}}{3} \right]$$

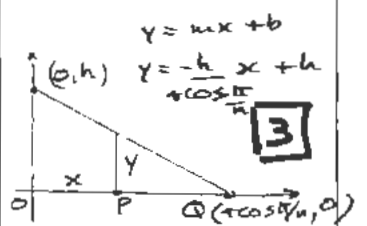
$$= 2h \tan \frac{\pi}{n} \times \frac{r^2 \cos^2 \frac{\pi}{n}}{6} = 2h \frac{\sin \frac{\pi}{n}}{\cos \frac{\pi}{n}} \times \frac{r^2 \cos^2 \frac{\pi}{n}}{6}$$

$$= \frac{r^2 h}{6} \cdot 2 \sin \frac{\pi}{n} \cdot \cos \frac{\pi}{n} = \frac{1}{6} r^2 h \sin \frac{2\pi}{n} \quad \text{q.e.d.}$$

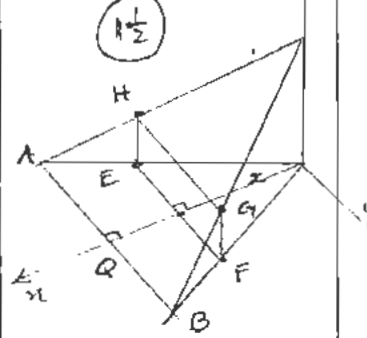
1 $\left(\frac{1}{2} \right)$

$\frac{1}{2}$

$\frac{1}{2}$



$\frac{1}{2}$



$$OQ = r \cos \frac{\pi}{n}$$

4

show/indicate/state evidence use of double $\angle$ s
 $2 \sin \theta \cos \theta = \sin 2\theta$

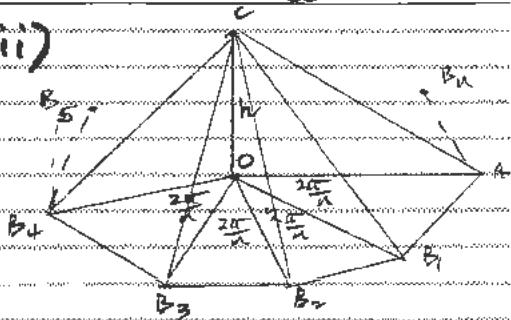
MATHEMATICS Extension 2: Question 4

Suggested Solutions

Marks

Marker's Comments

(b) (iii)



Volume of n wedges/pyramids

$$V_n = n \times \frac{1}{6} r^2 h \sin \frac{2\pi}{n}$$

$$\text{ie } V_n = \frac{1}{6} r^2 h n \sin \frac{2\pi}{n}$$

$$\therefore V = \lim_{n \rightarrow \infty} \frac{1}{6} r^2 h n \sin \frac{2\pi}{n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{6} r^2 h n \times \frac{\sin(\frac{2\pi}{n})}{(\frac{2\pi}{n})} \times (\frac{2\pi}{n})$$

$$\lim_{n \rightarrow \infty} \frac{1}{6} r^2 h \times 2\pi \times \frac{\sin(\frac{2\pi}{n})}{(\frac{2\pi}{n})}$$

$$= \frac{1}{6} r^2 h \times 2\pi \times 1$$

$$V = \frac{1}{2} \pi r^2 h$$

$$\text{let } x = \frac{2\pi}{n}$$

$$\therefore n = \frac{2\pi}{x}$$

$$\text{as } n \rightarrow \infty \quad x \rightarrow 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{6} r^2 h \cdot \frac{2\pi}{x} \cdot \sin x$$

$$\lim_{x \rightarrow 0} \frac{\pi r^2 h \sin x}{3x}$$

$$\therefore V = \frac{\pi}{3} r^2 h \times 1$$

$$V = \frac{1}{2} \pi r^2 h$$

"For small angles"

$$\text{as } n \rightarrow \infty \quad \frac{2\pi}{n} \rightarrow 0$$

$$\therefore \sin \frac{2\pi}{n} \div \frac{2\pi}{n}$$

$$\therefore \frac{1}{6} r^2 h n \cdot \sin \frac{2\pi}{n} \div \frac{1}{6} r^2 h n \cdot \frac{2\pi}{n}$$

$$\therefore V_n \div \frac{1}{2} \pi r^2 h \quad \text{as } n \rightarrow \infty$$

$$\therefore V = \lim_{n \rightarrow \infty} V_n = \frac{1}{2} \pi r^2 h$$

$\frac{1}{2}$

2

For organising clearly + correctly

$\frac{1}{2}$

$$? \sin \frac{2\pi}{n} \rightarrow \frac{2\pi}{n}$$

$\frac{1}{2}$